

Ahmed Ahmed

R&D Power & Embedded Electronics Engineer

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PROFESSIONAL SUMMARY

R&D engineer with 4 years delivering embedded power systems, motor-driver architectures, and advanced PCBs for UAV and robotics applications. Proven track record across BLDC/ESC development, battery and BMS design, 200W wireless power transfer, STM32 firmware on FreeRTOS, and high-density KiCad PCB layout – from concept through hardware validation, with the ability to interpret academic and technical literature and translate research concepts into functional engineering prototypes.

Hands-on experience in government-funded R&D (TUBITAK), academic research, electric-vehicle power systems, and industrial embedded platforms. Strong fit for Revotronics Labs combining motor-driver design, power-conversion expertise, real-time firmware development, and rigorous prototype-to-system delivery in demanding R&D environments.

TECHNICAL SKILLS

- **Power Electronics:** Motor drivers, BLDC motors, ESCs, FOC, Space Vector PWM, Six-Step Commutation, PID control, buck/boost converters, gate drivers, SMPS, BMS, battery charging, wireless power transfer (200W WPT), galvanic isolation, high-current PCB design, power distribution, analog systems, ADC/DAC interfacing, IC selection, signal filtering, EMC considerations.
- **Embedded Firmware:** Embedded C/C++, bare-metal programming, FreeRTOS, FreeRTOS-MPU, FreeRTOS+TCP, deterministic task management, fault monitoring, real-time safety logic, DMA, UART, SPI, I2C, CAN, Modbus, SMBus, Ethernet, TCP/IP, SD-card data logging.
- **Microcontrollers & Platforms:** STM32H7, STM32F7, STM32, RP2040, ESP32, Arduino, NEXTION HMI, ToF sensors, Multi-IMU sensor arrays.
- **PCB Design & Hardware:** Schematic design, advanced KiCad PCB layout, Altium Designer, Eagle, KiCad plugin development, high-density LED PCB, power-management PCB design, oscilloscope, function generator, soldering, hardware validation and debugging.
- **UAV, Robotics & Control:** UAV electronics, drone wireless charging, flight controller architecture, ArduPilot/PX4 analysis, robotics control systems, GPS-denied localization, Kalman Filtering, sensor fusion, ROS, embedded Linux (3+ years daily use).
- **Programming Languages:** C, C++, C#, Python, MicroPython, JavaScript.
- **Languages:** Arabic (native), Turkish (advanced – C1), English (advanced).

EXPERIENCE

Zenopix Technology <i>R&D Embedded Systems Engineer</i>	Konya, Turkey 2021–Present
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UAV Wireless Charging Station	2026–Present (Ongoing)
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- Architected an integrated wireless charging platform for UAV applications, engineering ground transmitter (Tx) and onboard receiver (Rx) power electronics for autonomous drone charging cycles.
- Developed a custom BMS architecture for UAV battery packs covering safe charging profiles, real-time battery status monitoring, battery-health optimization, and embedded fault handling via SMBus communication.
- Integrated power conversion, BMS, and embedded control requirements into an R&D platform aligned with UAV wireless power transfer and firmware development objectives.

Industrial BLDC Motor Driver	2026–Present (Ongoing)
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- Defined the architectural framework for a robust industrial BLDC motor driver optimized for high-reliability operation in demanding electromechanical systems.
- Formulated motor-control design requirements covering drive stages, control-loop stability, protection logic, thermal robustness, gate driver selection, and firmware interfaces.
- Structured the hardware concept around FOC and Six-Step control, high-current PCB layout, buck/boost-stage considerations, and hardware validation planning.

Hologram Fan System – TUBITAK R&D Project

2024–2026

- Engineered a custom 0.5-meter high-density LED PCB for a Persistence of Vision display, selecting high-performance LEDs and driver ICs capable of high-frequency switching.
- Designed high-current KiCad PCBs with strict galvanic isolation between power electronics and high-speed control circuitry for safer operation and cleaner signal behavior.
- Implemented BLDC motor-control algorithms including FOC, Space Vector PWM, Six-Step Commutation, and PID loops to regulate speed stability and display synchronization.
- Architected STM32 firmware with FreeRTOS for deterministic task management, real-time safety monitoring, fault detection, error handling, and system recovery.
- Developed a resonance-based wireless power transfer system by designing customized Tx/Rx circuits and achieving continuous 200W power transmission.
- Applied Fuzzy Logic Control to dynamically regulate wireless power delivery and improve transfer stability under varying operating conditions.
- Validated power electronics, PCB assemblies, and embedded control behavior using oscilloscopes, function generators, rapid prototyping, soldering, and hands-on hardware debugging.

End-to-End Control System – Abrasion Tester Machine

2021–2023

- Built the complete signal-processing pipeline for accurate sensor-data acquisition using digital filtering, analog filtering, and embedded measurement logic over SPI and I2C interfaces.
- Developed a custom DC motor driver with PID algorithms for accurate speed calibration and repeatable mechanical testing.
- Programmed a PC-based C# graphical user interface to control the machine, monitor operating behavior, and log test data in real time via UART communication.
- Delivered the control-board hardware and firmware from concept through debugging, calibration, and operational validation.

Embedded Interface Development & UI Simulation

2021–2022

- Developed an interactive JavaScript/CSS lighting simulation demonstrating environmental lighting effects for customer-facing product visualization.
- Launched four localized multilingual corporate websites and implemented SEO optimizations to improve online discoverability.

PROJECTS

Indoor Unmanned Vehicle Localization for Drift Reduction Using Multi-IMU Sensor Fusion

2025–Present

Master's Thesis | Multi-IMU, Sensor Fusion, STM32H7, FreeRTOS, Kalman Filtering

- Built a high-performance embedded system using STM32H7 to process approximately 250 raw data channels from a custom Multi-IMU hardware array in real time over SPI and I2C buses.
- Implemented FreeRTOS, FreeRTOS-MPU, and FreeRTOS+TCP for deterministic multitasking, memory protection, robust network communication, and scalable firmware architecture.
- Developed Kalman Filtering and sensor-fusion algorithms to estimate navigation states and reduce drift in GPS-denied UAV and robotics environments.
- Integrated DMA-driven data acquisition, Ethernet communication, and real-time processing into a research platform applicable to autonomous vehicles and embedded control.

Flight Controller Architecture & Prototyping – ArduPilot / PX4

2025–Present

Project | ArduPilot, PX4, Flight Controller, PCB Design, Sensor Isolation

- Conducted architectural analysis and reverse engineering of ArduPilot and PX4 flight stacks to define hardware, firmware, and communication protocol requirements for a custom UAV flight controller.
- Designed the first iteration of a custom flight-controller PCB with attention to sensor isolation, power distribution, component placement, and KiCad layout constraints.
- Formulated a bare-metal firmware strategy around control loops, sensor fusion, embedded safety behavior, and future UAV ecosystem compatibility.

Laser Vibrometer PCB & System Design

2024

Freelance | PCB Design, Power Management, Sensors, Data Logging

- Designed a custom PCB with battery charging, power-management circuitry, precision laser control, and embedded measurement interfaces over I2C and UART.
- Integrated ToF sensor for accurate distance and vibration measurement in an embedded instrumentation platform.
- Developed PC communication protocols and high-speed SD-card data logging to support repeatable test capture and analysis.

Temperature Control System – Cosmetic Medical Device Furnace

2023

Freelance | Embedded Systems, PCB Design, PID Control

- Designed the hardware layout and primary control PCB for a specialized medical thermal-control device.
- Implemented PID control algorithms to maintain accurate and stable furnace temperature regulation.
- Programmed an interactive NEXTION HMI interface for operator control and real-time monitoring via UART communication.

International Efficiency Challenge Electric Vehicle Races – Electromobile Category

2024–2025

Competition | Motor Driver, Power Supply, BMS, SMPS, EV

- Designed a motor driver for a 4.5 kW, 72V BLDC motor used in an electric vehicle competition platform.
- Designed a 3 kW power supply based on a 220V-to-72V AC transformer for induction-load inspection tests.
- Researched and supported Battery Management System (BMS) development for electric-vehicle energy storage and protection via CAN communication.
- Contributed to the design of a local charging unit applying SMPS topologies and power-conversion principles.
- Coordinated project delivery, technical execution, and team performance under competition deadlines.

Hyperloop Development Competition

2023

Competition | STM32F7, Ethernet, TCP/IP, Embedded Networking

- Designed a data-transmission system between the train and monitoring station using STM32F7 Ethernet capability and a TCP/IP stack.
- Deployed a TCP-based communication path through router and Wi-Fi infrastructure for computer-side monitoring.
- Validated embedded network communication concepts applicable to high-speed robotics and autonomous transportation systems.

POSYAK Medical Assistive Device – Barrier-Free Living Technologies

2022

Competition | Wearable, Motion Sensing, Mobile App

- Developed a wearable assistive device for Parkinson's patients that generated posture alerts when neck remained bent beyond a defined duration threshold.
- Integrated I2C-based IMU motion-data collection with a mobile application to record and process head-movement behavior.

Additional Engineering Competitions and Early Technical Projects

2014–2024

Projects | UAV, Robotics, Embedded Systems, Education

- Contributed to a Fighting UAV Competition project, participating in unmanned aerial vehicle design and development tasks.
- Contributed to an Agricultural Unmanned Ground Vehicle Competition focused on autonomous ground-vehicle functionality for agricultural applications.
- Designed and built a rotary-wing UAV for the International UAV Competition with emphasis on agility, reliability, and embedded control efficiency.
- Guided younger participants as a technical consultant in a Smart Transportation Competition, supporting concept development and engineering execution.
- Built educational-technology and quiz software using C#, database integration, authentication screens, scoring logic, and user-interface design.
- Achieved first place in a middle-school electronics competition by accurately assembling a random schematic circuit in the shortest time.

TEACHING & COMMUNITY TRAINING

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- Delivered a student-association presentation on embedded systems, explaining architecture, applications, and practical design considerations.
 - Developed and launched an online course on electronics fundamentals to help learners build a strong practical foundation.
 - Mentored students and children through technical training sessions in electronics and embedded systems, encouraging hands-on learning and project-based practice.

EDUCATION

Necmettin Erbakan University Master's Degree, Mechatronics Engineering	Konya, Turkey 2024–Present
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Necmettin Erbakan University Bachelor's Degree, Mechatronics Engineering GPA: 3.06 / 4.00 <i>Graduated with Honours (Onur Belgesi)</i>	Konya, Turkey 2020–2024
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TOMER Turkish Language Course, C1 Level	Istanbul, Turkey 2019–2020
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AL KINDY High School High School, Natural Sciences	Kuwait 2016–2019
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CERTIFICATIONS & AWARDS

- International Efficiency Challenge Electric Vehicle – Teknofest Finalist, 2025.
- International Efficiency Challenge Electric Vehicle – Teknofest Finalist, 2024.
- Graduation with Honours (Onur Belgesi) – Necmettin Erbakan University, 2024.
- Microcontroller Embedded C Programming: Absolute Beginners – Udemy, 2023.
- Mastering RTOS: Hands-on FreeRTOS and STM32Fx with Debugging – Udemy, 2023.
- Linux Training (Ubuntu) – Komek, 2022.
- First Aid Certification – Komek, 2022.
- Arduino Programming and Development Course – Komek, 2021.